

ExSOIL Container Infrastructure: Presentation Walkthrough

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A suggested flow for walking someone through the project's progress. Assumes the audience is not deeply familiar with the CESM/CTSM software stack. Total time: 10-15 minutes plus questions.

1. The goal (1 minute)

Start with what the project does, not how it works.

Key points:

- The ExSOIL project runs land surface simulations at real ecological observatory sites (NEON network, 48 sites across the U.S.) and compares the model's predictions against what the tower instruments actually measured.
- The variables we care about: soil temperature, soil moisture, carbon fluxes, sensible heat flux. These are what tell us whether the model's representation of soil processes is accurate.
- The computational environment is delivered as a Docker container. Researchers pull the container, launch JupyterLab, and work with simulations and data. No installation, no configuration, no dependency headaches.

2. Where we started (1-2 minutes)

Explain the problem we inherited.

Key points:

- The previous container (`escomp/cesm-lab-neon`, last updated October 2022) was built on end-of-life software: CentOS 8, Python 3.7, Intel-only binaries.
- Team members on Apple Silicon Macs (M1-M4) could not run it reliably. It either crashed or ran under processor emulation (slow, unstable).
- The NEON tower site workflow (the core research capability) was non-functional. It had been assembled from an undocumented mix of component versions that no longer worked together.
- In short: the container could not run on modern hardware, and even on compatible hardware, it could not perform its primary research function.

3. How the pieces fit together (2-3 minutes)

Use the architecture guide and lineage chart to orient the audience before describing what changed.

Open: `docs/ctsm-architecture-guide/lineage-chart.png`

Key points:

- CESM is the full Earth system model (atmosphere, ocean, ice, land). It is very large and we only need the land component.
- CTSM is the standalone land model. It includes CLM (the land simulation code), CIME (the build/case management system), and the NEON tower workflow. CTSM is what NCAR recommends for single-site land experiments.
- The old container used the full CESM and grafted in NEON support from a development branch. That graft broke as the components diverged.
- Our rebuild uses standalone CTSM 5.4, which includes NEON natively. This aligns with NCAR's intended approach and eliminates the custom graft.

If questions arise about CESM vs CTSM: docs/ctsm-architecture-guide/ctsm-architecture-guide.html has four diagrams covering the component hierarchy, build pipeline, data flow, and NEON workflow.

4. What we built (3-4 minutes)

Walk through the deliverables. Concrete outcomes, not process.

Multi-platform container:

- Runs natively on both Intel/AMD and Apple Silicon (arm64). Docker selects the right architecture automatically.
- Ubuntu 24.04 base (supported through 2029), Python 3.13, current scientific libraries.
- All compiled dependencies (MPICH, HDF5, NetCDF, compilers) come from conda-forge pre-built binaries. Build time dropped from ~45 minutes to ~5 minutes.
- Exact package versions pinned via conda-lock for reproducibility across platforms.

NEON simulation pipeline (end-to-end):

- 48 NEON tower sites discoverable and configurable.
- The model compiles from Fortran source inside the container (~100 seconds).
- A 1-day transient simulation at KONZ (Konza Prairie, Kansas) runs to completion and produces valid CLM output: 31 variables, 48 half-hourly time steps.
- Output includes the core research variables: soil temperature (TSOI), soil moisture (H2OSOI), sensible heat flux (FSH).
- Output is readable directly with xarray for analysis in Python.

Validation:

- 90-test automated suite across three tiers: environment checks, case management workflow, full model compilation and analysis.
- All 90 tests pass on native arm64.

Documentation:

- 6 Architecture Decision Records (why each technical choice was made)
- Architecture guide with diagrams (for domain experts, not just engineers)
- Decision trail documenting the path from the initial problem through each resolution
- NSF progress report, roadmap, changelog, getting-started guide

5. Challenges worth mentioning (1-2 minutes)

These give context for why it took the effort it did.

The NEON gap: NEON tower support was developed for standalone CTSM starting in 2021. It was never part of any CESM 2.x release and will not enter CESM until version 3.x (targeted late 2026). The previous container masked this by using an undocumented custom build. Our rebuild exposed the gap, which led to the architectural decision to move to standalone CTSM.

Data infrastructure migration: NCAR migrated their data servers between the CTSM 5.4.002 release (December 2025) and the current development branch. The release shipped before configuration files were updated to point at the new server. This was not documented in the release notes. We resolved it by upgrading to a newer CTSM tag (5.4.043) and building a pre-download script with retries and server fallback.

MPI compatibility: The NEON workflow defaults to a serial MPI mode, but conda-forge's MPICH libraries are always present. The two conflict at runtime. We patched the defaults to use real MPICH, which works correctly with the standard launcher.

6. What is next (1 minute)

Near-term:

- Extend validation to longer run periods and additional NEON sites
- Connect the analysis notebooks (Modeling_Hub, Design_Hub) to simulation output for model-data comparison workflows
- Cache input data on university infrastructure to eliminate the slow download step for new users

Medium-term:

- Merge to main and publish the updated container image
- CI/CD integration for automated testing on both architectures
- Evaluate multi-site batch runs for systematic model evaluation

Reference documents to have open

Document	When to use it
lineage-chart.png	Section 3: explaining how components relate
ctsm-architecture-guide.html	If architecture questions go deeper
plain-language-explanation.md	If data availability questions come up
infrastructure-progress-report.md	Leave-behind: formal summary of all work
ROADMAP.md	“What’s next” questions

If asked: “Can you show me?”

The most compelling demonstration is the Getting Started notebook inside the running container:

```
docker run --rm -p 8888:8888 exsoil-ctsm543-test
```

Open the URL in a browser, then open `notebooks/Getting_Started_CTSM_NEON.ipynb`. This shows NEON site discovery, case configuration, and environment verification without needing to run a full simulation (which takes ~15 minutes for data download and build).

For a full simulation demo, run the pre-download script first (the slow step), then the notebook can create and build a case interactively.